Non-Product Kamke Δ-Functions and Redundancy of the Initial Derivative Clause

Automated Math Discovery Candidate Proof

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Status. Candidate full answer to the Kamke Δ-function question in the source paper, likely valid pending expert review. A broad polynomial family supplies non-product examples. More strongly, an example for which the initial Δ-derivative requirement is essential cannot exist under the stated axioms.

Source and Question

This packet concerns the final paragraph on Kamke Δ-functions in Section 5, page 29, of Oberta [1], arXiv:2512.13602v2. Definition 7 there calls

$$w : [a,b]_{\mathbb{T}} \times [0,\infty) \longrightarrow [0,\infty)$$

a Kamke Δ-function when:

1. $w(\cdot, x)$ is rd-continuous for every $x \geq 0$;
2. the family $\{w(t,\cdot) : t \in [a,b]_{\mathbb{T}}\}$ is uniformly equicontinuous on every bounded interval $[0, R]$;
3. $w(t, 0) = 0$;
4. zero is the unique nonnegative continuous solution of

$$u(t) \leq \int_a^t w(s, u(s))\, \Delta s \tag{1}$$

among those u that are Δ-differentiable at a and satisfy $u^{\Delta}(a) = 0$.

The paper asks for more sophisticated examples, for instance one not of the form $q(t)h(x)$, or one having a nonzero continuous solution of (1) that is excluded only by the derivative requirement.

Theorem 6 is an extension of Theorem 11.2 in [4] to the case of an arbitrary time scale. However, note that in [4], the proof of the first 5 axioms of Definition 5 is skipped. Thus, the proof presented by us supplements also the case of $\mathbb{T}=\mathbb{R}$ proved in [4], as we provide the proofs for these axioms as well. Also, note that in the proof of Theorem 6, verification of Axiom (vi) of Definition 5 follows precisely the above-mentioned proof from [4].

The notion of a Kamke Δ-function (see Definition 7) is a new concept in the theory of time scales. It extends the notion of a Kamke function presented in Definition 3.1 in [15] and Section 13.3 in [4], to the case of an arbitrary time scale. In particular, note that for $\mathbb{T}=\mathbb{R}$, Kamke Δ-function reduces to these two above mentioned notions of Kamke function. In Theorem 10 we provide a particular example of a Kamke Δ-function, which for $\mathbb{T}=\mathbb{R}$ reduces to the well-known example of a Kamke function usually found in the literature. However, it might be an interesting question to find more sophisticated examples of Kamke Δ-functions, e.g. one which is not of the form $q(t)\cdot h(x)$, or one for which $u\equiv 0$ is not the unique non-negative continuous solution of (2.46) (and thus the additional requirement regarding Δ-differentiability at a from Axiom (iv) of Definition 7 would need to be employed).

Theorem 11 was inspired by Section 3 in [5] and Theorem 17 in [18], where similar systems of classical differential equations and fractional differential equations, respectively, are studied. Also, note that Section 4.3 was inspired by Section 4.3 in [18], where a similar fractional parabolic PDE (for $\mathbb{T}=\mathbb{R}$) is investigated. Moreover, note that a more general non-linearity, e.g. in the form of a p-Laplacian, could be considered in Example 5 in the

Proof Intuition

There are two independent observations. First, the derivative condition in Definition 7 is already forced by the integral inequality. The inequality at $t=a$ gives $u(a)=0$. If a is right-scattered, the first Δ-integral step is $\mu(a)w(a,u(a))=0$, so $u(\sigma(a))=0$. If a is right-dense, uniform equicontinuity of all the functions $w(t,\cdot)$ at zero makes the integrand uniformly small on the shrinking interval $[a,t]_{\mathbb{T}}$. Dividing by $t-a$ then shows $u(t)/(t-a)\to 0$. Thus the extra derivative condition can never separate a nonzero solution from zero.

Second, many non-product examples are nevertheless easy to build. Any finite polynomial in x with bounded nonnegative rd-continuous coefficients is at most linear in u on the bounded range of a candidate solution. Time-scale Gronwall therefore forces that solution to vanish. Choosing nonproportional coefficient functions prevents a factorization $q(t)h(x)$.

Automatic Endpoint Derivative

We use the standard notation σ for the forward jump and $\mu(t)=\sigma(t)-t$ for graininess. As in the source paper, the compact interval $[a,b]_{\mathbb{T}}$ is treated as the ambient time scale and $a<b$.

Theorem 1 (The derivative clause is automatic)**.** *Suppose w satisfies axioms (i)–(iii) of Definition 7 in the source paper. If a nonnegative continuous function $u:[a,b]_{\mathbb{T}}\to[0,\infty)$ satisfies* (1) *for every $t\in[a,b]_{\mathbb{T}}$, then u is Δ-differentiable at a and*

$$u^{\Delta}(a)=0.$$

Proof. Putting $t=a$ in (1) gives $0\le u(a)\le 0$, hence $u(a)=0$.

Suppose first that a is right-scattered. The one-step identity for the Cauchy Δ-integral gives

$$0\le u(\sigma(a))\le\int_a^{\sigma(a)} w(s,u(s))\,\Delta s=\mu(a)w(a,u(a))=0.$$

Thus $u(\sigma(a)) = u(a) = 0$. Continuity at a right-scattered point implies Δ-differentiability, and the difference-quotient formula yields

$$u^{\Delta}(a) = \frac{u(\sigma(a)) - u(a)}{\mu(a)} = 0.$$

Suppose instead that a is right-dense. Fix $\varepsilon > 0$. Uniform equicontinuity in axiom (ii), applied at $x = 0$, and axiom (iii) give an $\eta > 0$ such that

$$0 \leq y < \eta \quad \Longrightarrow \quad 0 \leq w(s, y) < \varepsilon \qquad (s \in [a, b]_{\mathbb{T}}). \tag{2}$$

Since u is continuous at a and $u(a) = 0$, there is $\delta > 0$ such that $u(s) < \eta$ whenever $s \in [a, b]_{\mathbb{T}}$ and $0 \leq s - a < \delta$. Hence, for $t \in (a, a + \delta) \cap \mathbb{T}$, (1), (2), and $\int_a^t 1\,\Delta s = t - a$ imply

$$0 \leq \frac{u(t) - u(a)}{t - a} \leq \frac{1}{t - a} \int_a^t \varepsilon\,\Delta s = \varepsilon.$$

Therefore the time-scale limit of the difference quotient at a is zero, which is precisely $u^{\Delta}(a) = 0$ in the right-dense case. □

Corollary 1 (Equivalent definition and impossibility result)**.** *Under axioms (i)–(iii), axiom (iv) of Definition 7 is equivalent to the following apparently stronger statement:*

> $u \equiv 0$ *is the unique nonnegative continuous function satisfying* (1)*, with no endpoint differentiability assumption.*

Consequently, the source paper's proposed example in which a nonzero solution is excluded only by the condition $u^{\Delta}(a) = 0$ *does not exist.*

Proof. The unconditional uniqueness statement plainly implies axiom (iv). The reverse implication follows from Theorem 1, because every nonnegative continuous solution of (1) automatically belongs to the class quantified over in axiom (iv). □

A Family of Non-Product Examples

Proposition 1 (Polynomial Kamke Δ-functions)**.** *Let $m \geq 1$, and let $p_j \in C_{\mathrm{rd}}([a, b]_{\mathbb{T}}, [0, \infty))$ for $1 \leq j \leq m$. Then*

$$w(t, x) = \sum_{j=1}^{m} p_j(t) x^j \tag{3}$$

is a Kamke Δ-function.

If the coefficient vectors $(p_1(t), \ldots, p_m(t))$ are not all proportional as t varies, then (3) *cannot be written as $q(t)h(x)$.*

Proof. For fixed x, the function $w(\cdot, x)$ is rd-continuous, and $w(t, 0) = 0$. Every p_j is bounded on the compact time-scale interval. For a fixed $R > 0$, the mean-value theorem in the x variable gives, for $x, y \in [0, R]$,

$$|w(t, x) - w(t, y)| \leq \left(\sum_{j=1}^{m} j\,\|p_j\|_{\infty} R^{j-1} \right) |x - y|,$$

uniformly in t. Thus axioms (i)–(iii) hold.

Let u be a nonnegative continuous function satisfying (1), and put $M = \max_{[a,b]_{\mathbb{T}}} u$. For $0 \le x \le M$,

$$w(t,x) \le Q_M(t)x, \qquad Q_M(t) := \sum_{j=1}^{m} p_j(t)M^{j-1}.$$

The coefficient Q_M is nonnegative and rd-continuous. Therefore

$$u(t) \le \int_a^t Q_M(s)u(s)\,\Delta s.$$

The time-scale Gronwall inequality (the same result used in Theorem 10 of the source paper) gives $u(t) \le 0$ on $[a,b]_{\mathbb{T}}$; hence $u \equiv 0$. This proves axiom (iv), indeed in its unconditional form.

Finally, suppose $w(t,x) = q(t)h(x)$. At any time t_0 for which the coefficient vector is nonzero, $q(t_0) \ne 0$, and equality for all $x \ge 0$ forces h to be the polynomial $w(t_0,\cdot)/q(t_0)$. Comparing polynomial coefficients then shows

$$(p_1(t),\ldots,p_m(t)) = \frac{q(t)}{q(t_0)}(p_1(t_0),\ldots,p_m(t_0))$$

for every t. Hence all coefficient vectors would be proportional, a contradiction. □

Corollary 2 (Explicit non-product example)**.** *On every nondegenerate compact time-scale interval* $[a,b]_{\mathbb{T}}$,

$$\boxed{w(t,x) = x + (t-a)x^2}$$

is a Kamke Δ*-function and is not of the form* $q(t)h(x)$*.*

Proof. Apply Proposition 1 with $p_1(t) = 1$ and $p_2(t) = t - a$. For a direct factorization check, the 2×2 determinant using $t = a, b$ and $x = 1, 2$ is

$$\det\begin{pmatrix} w(a,1) & w(a,2) \\ w(b,1) & w(b,2) \end{pmatrix} = \det\begin{pmatrix} 1 & 2 \\ 1+b-a & 2+4(b-a) \end{pmatrix} = 2(b-a) \ne 0.$$

A product $q(t)h(x)$ would give a rank-one matrix and determinant zero. □

Verification, Scope, and Novelty

The endpoint proof was audited against the two exhaustive cases in Remark 17 of the source paper. Its only time-scale identities are the right-scattered one-step integral formula, the right-dense difference-quotient formula, and $\int_a^t 1\,\Delta s = t - a$, all stated in the source preliminaries. The polynomial example uses the same Gronwall theorem that Oberta invokes for the linear example in Theorem 10. No numerical or computer-assisted step enters the proof.

The result answers precisely the two example directions in the final Kamke-Δ paragraph: one direction positively by a large non-product family, and the other negatively by proving impossibility. It does not claim to modify the paper's main existence theorem or resolve its other concluding remarks.

A bounded novelty check on August 11, 2026 searched the run's registry, solution, attempt, and proof-gap indexes for arXiv:2512.13602 and the phrases "Kamke Δ-function," "non-product," and "endpoint Δ-differentiability." Exact-title and close-phrase web searches found no independent answer. OpenAlex listed one 2026 citing work, on fractional retarded dynamic equations, whose indexed title and abstract do not state either result here. Novelty confidence is therefore moderate, not definitive, pending a specialist literature review.

The recommended human-review focus is the quantifier use of uniform equicontinuity in (2), the endpoint convention for $[a, b]_{\mathbb{T}}$, and the application of time-scale Gronwall to Q_M.

References

[1] D. Oberta, *On the existence of solutions of dynamic equations on time scales in Banach spaces*, arXiv:2512.13602v2 (2026); published online in Math. Methods Appl. Sci., DOI: 10.1002/mma.70827.

[2] M. Bohner and A. Peterson, *Dynamic Equations on Time Scales: An Introduction with Applications*, Birkhäuser, Boston, 2001. See Theorem 6.4 for the Gronwall inequality used by the source paper and Proposition 1.